Prompt Language Doesn't Matter, but Keep the Original Text: Large Language Models and Sentiment Analysis in French News

Abstract

For a decade, comparative text analysis had one answer to non-English text, which was to translate it into English and apply English tools. Large language models (LLMs) inherit that reflex, and a second one, to prompt in English. This article tests both on French news. Fifteen open- and closed-weight models scored 200 hand-coded sentences under three workflows, a French prompt on the original, an English prompt on the original, and an English prompt on a machine translation. Prompt language made no difference. Equivalence tests rule out any substantial advantage for English, and the result holds on the full corpus of 2,683 articles. Translation did make a difference. The same machine translation that improved the Lexicoder dictionary lowered the LLMs' agreement with human coders, so the strategy that served English-bound tools becomes a cost for models that read the source language. Every model from about 8B parameters upward beat both the native French dictionary and the translated English one, reaching r = 0.81 against 0.36, for a few cents per thousand sentences and without any training data or fine-tuning. The language of the prompt is a matter of convenience. The language of the text is not: analyze it in the original.

Keywords: Sentiment Analysis, Large Language Models, Machine Translation, Prompt Language, French Language, Open-Weight Models, Computational Social Science

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Introduction

For ten years, the advice to a researcher facing a corpus in French, German, or Arabic was simple: translate it into English and use the English tools. Computational text analysis grew up around English, and its tools, from sentiment dictionaries to topic models, were built for English text. This dominance has been criticized for depressing comparative research and for excluding scholarly communities that work in other languages (Baden et al., 2022). Machine translation was the workaround, and it was validated repeatedly for dictionaries and topic models (De Vries et al., 2018; Lucas et al., 2015). Large language models (LLMs) change the terms of the problem. Used without task-specific training, they match or exceed crowd workers and expert coders on common annotation tasks (Gilardi et al., 2023; T"ornberg, 2025), and they do so, with some loss, in languages other than English (Heseltine & Clemm Von Hohenberg, 2024; Rathje et al., 2024). For the many researchers who work in a language other than English and have neither the labeled data nor the skills to train or fine-tune a model, this is the first tool that reads their texts as they are. This article shows that the old advice now points the wrong way.

Yet researchers who use LLMs on non-English text often keep two reflexes from the English-centered era. The first is to prompt in English. The justification is that these models are trained mostly on English (Touvron et al., 2023) and that English instructions have been found to work better even for non-English inputs (Lai et al., 2023; X. V. Lin et al., 2022). English prompts are also cheaper, because tokenizers trained on English-heavy data split French into more pieces (Ahia et al., 2023). The second is to translate the text into English before analysis. This practice comes from dictionaries and other bag-of-words methods, whose lexicons were English, and it reappears as "translate-test" with LLMs (Etxaniz et al., 2024). Whether either reflex helps, harms, or makes no difference is unsettled, and the social science evidence is thin.

This article tests both reflexes on sentiment analysis, a measurement task central to political communication research (van Atteveldt et al., 2021; Young & Soroka, 2012). Sentiment

is a demanding test for translation, because affective meaning is precisely what translation can alter (Mohammad et al., 2016). The benchmark is a set of 200 sentences drawn from 2,683 articles on open-source software published by 13 French-language dailies in Québec and France, each sentence scored by three human coders. Fifteen open- and closed-weight language models scored them with a French prompt, with an English prompt, and with an English prompt on a machine translation. The French and English Lexicoder dictionaries provide the comparison that most applied researchers face. Equivalence tests establish the absence of a prompt-language effect as a finding rather than a non-result, and the best model then rescored the full corpus of 2,683 articles to check that the result holds at scale.

The results speak to both reflexes. Prompt language made no meaningful difference, though the English prompt is cheaper. Translation did: the same machine translation that lifted the dictionary workflow from r = 0.36 to 0.55 lowered every model of at least 8B parameters, and the models beat both dictionaries without it. The article also answers the practical question that follows: is a researcher working on French text still better served by a dictionary? On this evidence, no. Every model from about 8B parameters upward outperformed both the native French dictionary and the translated English one on correlation with human judgments, open-weight models included, for a few cents per thousand sentences.

Literature Review

LLMs as Measurement Instruments in the Social Sciences

Zero-shot LLM annotation has moved quickly from curiosity to candidate standard. The paper that set the agenda had ChatGPT annotate 6,183 English tweets and news articles without any example. Its accuracy exceeded that of crowd workers by about 25 percentage points on average, its intercoder agreement exceeded that of both crowd workers and trained annotators on every task, and each annotation cost about thirty times less than paying a crowd worker (Gilardi et al., 2023). GPT-4 outperformed expert coders and supervised classifiers at identifying the party of politicians from their social media posts in eleven countries (T"ornberg, 2025). GPT-4 also closely approximated expert coding of negativity and sentiment in tweets from the United States,

Chile, Germany, and Italy (Heseltine & Clemm Von Hohenberg, 2024). These results have fed broad claims that generative AI could transform computational social science (Ziems et al., 2024), chiefly by removing the need for the thousands of annotated examples that supervised learning requires (Laurer et al., 2024).

A more cautious literature qualifies these claims on three fronts. First, performance is uneven. Across 25 English benchmarks, zero-shot LLMs did not outperform the best fine-tuned models on classification tasks (Ziems et al., 2024). When GPT-4 replicated 27 annotation tasks from 11 datasets in published social science articles, its performance varied with the dataset and with the kind of task, and on a third of the tasks the model missed or mislabeled at least half of the relevant cases (Pangakis et al., 2023). Validation against human coding therefore remains indispensable (Grimmer & Stewart, 2013; H. Lin & Zhang, 2026), and it is only as informative as the human gold standard is reliable (Song et al., 2020). Second, established tools retain advantages. Dictionaries generalize across domains and preserve gradations of sentiment (Van Der Veen & Bleich, 2025). However, off-the-shelf dictionaries validate poorly against human judgment (van Atteveldt et al., 2021), and transformer-based models outperform them in German political and cultural texts (Klähn et al., 2025; Widmann & Wich, 2023). A domain-specific multilingual classifier can also match GPT-4o at a fraction of the cost (Mochtak et al., 2025). Third, reliance on proprietary models threatens reproducibility (Spirling, 2023). This has prompted calls to justify their use (Palmer et al., 2023) and evidence that open-weight models can be competitive alternatives (Klähn et al., 2025).

These debates leave the question of language largely implicit. The broadest evaluations rely on English benchmarks (Ziems et al., 2024). Studies that extend to other languages report slightly lower performance on non-English text (Heseltine & Clemm Von Hohenberg, 2024), or lower accuracy outside the United States that the authors attribute to party systems rather than to language (T"ornberg, 2025). Neither examines how researchers should interact with models in those languages. The most extensive multilingual evaluation in the social sciences covers 12 languages, yet it compared GPT with dictionary methods on English headlines only (Rathje et al.,

2024). Researchers working in other languages must make two choices: the language of the prompt and the language of the text.

The Language of the Prompt

The case for prompting in English starts from two observations. LLMs are trained predominantly on English (Touvron et al., 2023), and they perform less well in other languages, especially lower-resourced ones (Ahuja et al., 2023; Lai et al., 2023). A mechanistic account suggests why. Llama-2 models seem to think in English before answering in another language. Given a French prompt, the model works out the meaning in its middle layers, and at that stage its internal state already points to the English word for the answer. Only in the last layers does it switch to the target language (Wendler et al., 2024). If models effectively reason through English, instructions written in English should be understood best. Several findings support this pivot-language view. ChatGPT performed better with English prompts even when the task and input were in other languages (Lai et al., 2023). English templates remedied the unexpectedly low accuracy of prompts handcrafted by native speakers (X. V. Lin et al., 2022). Models fine-tuned on English prompts performed best with English prompts even on non-English samples (Muennighoff et al., 2023). English instructions outperformed Arabic ones on average across eleven Arabic classification tasks (Kmainasi et al., 2025).

A competing view emphasizes multilingual competence. Large models show strong multilingual reasoning abilities even in underrepresented languages (Shi et al., 2022). For capable models, the language of the prompt appears to matter little. In the Arabic study just cited, differences across prompt languages were minimal for the strongest model, GPT-4o (Kmainasi et al., 2025). GPT-4 Turbo's labels for Indonesian tweets under English and Indonesian prompts agreed almost perfectly (Rathje et al., 2024). The evidence on both sides comes mostly from reasoning and classification benchmarks, older models, and lower-resourced languages. None of it tests whether prompt language affects the accuracy of a graded sentiment judgment in a well-resourced language. The two views imply competing expectations:

H1a (pivot language). English prompts yield more accurate sentiment measurement

of French text than French prompts.

H1b (multilingual competence). English and French prompts yield equivalent performance on French text.

The Language of the Text

The second reflex predates LLMs. The bridge strategy of translating multilingual corpora into English so that English-language tools can be applied (Lucas et al., 2015) has been validated repeatedly. Machine-translated and human-translated texts yield highly similar term-document matrices and topic models (De Vries et al., 2018). A machine translation of the Lexicoder dictionary captured legislative conflict across European parliaments (Proksch et al., 2019). For sentiment specifically, analysis of English translations of Arabic posts was competitive with analysis of the originals (Mohammad et al., 2016). The logic has carried into the LLM era. Translating inputs into English before inference improves the performance of multilingual models (Etxaniz et al., 2024). It is often hard to beat for lower-resourced languages written in non-Latin scripts (Ahuja et al., 2023).

The opposing view treats translation as a source of loss. Translation alters sentiment: affective words disappear or are replaced by words of opposite valence, ambiguous words are mistranslated, and culturally specific expressions lose their meaning (Mohammad et al., 2016). Approaches that avoid translation, such as multilingual sentence embeddings, yield comparable or more reliable results in some settings (Licht, 2023). Crucially, the case for translation was built largely on tools that cannot read the source language, such as English dictionaries and bag-of-words models. LLMs can. For them, the supporting evidence comes from benchmarks whose non-English versions were themselves created by translation, a setting that may favor translation-based methods, as the authors note (Etxaniz et al., 2024). It does not come from text written natively in the source language. Again, the two views imply competing expectations:

H2a (translation as bridge). Analyzing English machine translations yields sentiment measurement at least as accurate as analyzing the French original.

H2b (translation as loss). Analyzing English machine translations yields less accurate sentiment measurement than analyzing the French original.

Research Questions

These debates motivate two research questions. **RQ1:** Does the language used to interact with the model affect LLM sentiment measurement of French text? This covers the language of the prompt and the language of the text through machine translation. H1a–b and H2a–b offer competing answers. **RQ2:** Can general-purpose LLMs, open- and closed-weight,[1] measure sentiment in French text well enough, without fine-tuning, to be a viable alternative to dictionary-based methods? RQ2 is descriptive. The hypotheses were derived from the literature and were not preregistered.

Data & Methods

Data Collection and Preparation

The corpus consists of 2,683 French-language news articles from 13 media sources (9 from Québec, 1 from Ontario, 3 from France). It is an exhaustive collection of the articles discussing open-source software in the Eureka database's "Major daily newspapers (FR)" category. Articles were selected with the search terms in Table 1, joined by OR, between January 1, 1991 and January 1, 2025. The start date is the release year of the Linux kernel, the project that made open-source software a public topic, so the window covers the entire history of the subject in the press. In practice it excludes nothing: the earliest match dates from June 1995.

Table 2 shows the distribution of articles by source media. Articles were downloaded in HTML format with a Python/Selenium scraper, and publication metadata and full text were extracted with custom R functions.

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[1] "Open source" traditionally implies access aligned with free and open-source software principles. In the LLM context, the term usually refers to the public release of model weights. "Open-weight" describes this condition more accurately and is used throughout.

Table 1
Search Terms Used to Select Articles in the Eureka Database

Group	Search terms
French terms	*logiciel libre*, *logiciels libres*, *logiciel open source*, *logiciels open source*, *code source ouvert*, *code source libre*
English and borrowed terms	*open source*, *open-source*, *free software*, *software libre*
Proper names	*Free Software Foundation*, *Richard Stallman*

Table 2
Collected Articles by Source Media

Source	Articles	First article	Last article
Libération	769	1995-06-23	2024-11-04
Monde, Le	666	2001-01-09	2024-12-26
Figaro, Le	402	1998-10-21	2024-12-04
Devoir, Le	318	1998-07-20	2024-10-26
Soleil, Le (Québec, QC)	179	1998-11-08	2023-11-18
Presse, La	152	1998-03-26	2017-10-14
Journal de Montréal, Le	65	2006-12-20	2023-04-24
Journal de Québec, Le	45	2007-03-24	2024-10-19
Tribune, La (Sherbrooke, QC)	43	2000-03-29	2022-12-03
Droit, Le (Ottawa, ON)	17	1998-11-10	2014-07-17
Nouvelliste, Le (Trois-Rivières, QC)	13	2000-01-13	2022-03-05
Quotidien, Le (Saguenay, QC)	10	2008-11-01	2019-04-10
Voix de l'Est, La (Granby, QC)	4	2010-09-17	2015-02-14

Translation

The EN→EN condition described below reproduces the translate-then-analyze strategy reviewed above. For this condition, the French corpus was machine-translated into English with Google Translate's document translation service. An automated R workflow processed the documents in batches and preserved document IDs, so that the French and English versions remain aligned.[2]

[2] See `src/21_translate_to_english.R` in the project repository.

Manual Annotation

A validation benchmark of 200 sentences was randomly sampled from the corpus. Each sentence was machine-translated into English with Google Translate through the `polyglotR` package. This sample size falls within the 100 to 300 units generally recommended for validating automated sentiment measures against a manually coded gold standard (van Atteveldt et al., 2021). It yields 95% confidence intervals of roughly ±0.05 around a correlation of 0.80. Three independent coders rated the sentiment of each sentence on a scale from -1 (very negative) to 1 (very positive), using a custom Shiny application. The three ratings were averaged. This average is the ground truth against which both dictionary and LLM methods are evaluated.

Agreement between individual coders was moderate. Krippendorff's alpha (interval) was 0.597 (95% bootstrap CI [0.516, 0.656]). The mean pairwise correlation between coders was 0.687. All three coders assigned the same sign (negative, zero, or positive) to only 42.5% of sentences. Two features of the task account for much of this disagreement. Coders used a fine-grained scale with more than 60 distinct values, and most sentences carry mild sentiment, so small differences often cross zero. The ground truth averages three coders and is therefore more reliable than any single coder: the Spearman-Brown projection from the mean pairwise correlation puts the reliability of the three-coder mean at 0.868.

Dictionary-Based Sentiment Analysis

Lexicon-based scoring was applied to the French original and to the English translation as a classical baseline. For French text, a customized version of the French Lexicoder Sentiment Dictionary (`frlsd`) (Duval & Pétry, 2016) was applied to lowercased, stopword-removed, sentence-segmented text. Each sentence was scored as (positive terms − negative terms) / total words. For the English translation, the standard Lexicoder Sentiment Dictionary (Young & Soroka, 2012) was used, with additional handling of contractions, proper nouns, and negation (for example, "not good" scored as negative). Each sentence was scored as ((positive + negated negative) − (negative + negated positive)) / total words. The two dictionaries therefore differ in construction as well as in language.

LLM-Based Sentiment Analysis

Fifteen general-purpose language models were evaluated. They were selected for prominence and affordability, as realistic choices for researchers who need to analyze large corpora. Three closed-weight models from the major vendors are included as a reference point, the justification that Palmer et al. (2023) ask for: the studies that established LLM annotation in the social sciences relied on proprietary GPT models (Gilardi et al., 2023; Heseltine & Clemm Von Hohenberg, 2024; Rathje et al., 2024; T"ornberg, 2025). Three further models test specific claims. Mistral AI, a French developer, describes its models as particularly strong in French. It also reports that its Tekken tokenizer compresses French text about 30% more efficiently than its earlier tokenizer (Mistral AI, 2024). Two Mistral models test both claims.

The ten open-weight models are Llama 3.2 1B, Llama 3.2 3B, Llama 3.1 8B, GPT-OSS 20B, Mistral Small 3.2 24B, Qwen3 32B, Llama 4 Scout, GPT-OSS 120B, Qwen3 235B-A22B, and DeepSeek V3.2. They range from 1B to 671B total parameters. The five closed-weight models are Claude Haiku 4.5 (Anthropic), Gemini 3.5 Flash (Google), GPT-5.6 Luna (OpenAI), Mistral Large 4 (Mistral AI), and Jev 1.13 (TypeSafe). Mistral Large 4 was evaluated in its October 2026 preview release and is classified as closed-weight because its weights were not yet published. Jev tests a different kind of model. It is not a text generator: it answers a typed question with a probability for each possible answer. It received the same task as the other models, posed as a question whose five answer levels are the five anchors of the sentiment scale, worded as in the prompts below and in the language of each condition. Its score is the probability-weighted average of the five levels, mapped onto the -1 to 1 scale. This lineup updates an initial 2024-2025 evaluation of 11 comparable models with an identical design. The smallest model, Llama 3.2 1B, rarely followed the instruction to answer with a number: about one call in 23 returned a parseable score, and even after retries it scored only 58 to 74 of the 200 sentences, depending on the condition. It was retained to mark the lower end of the size range, but its metrics rest on a reduced and non-random subset of sentences and are reported separately.

Each model was tested under three conditions: French prompts on French text (FR→FR),

English prompts on French text (EN→FR), and English prompts on the machine-translated English text (EN→EN).

All prompts followed the same structure: a task definition, a five-point sentiment scale from -1.0 to 1.0 with anchor descriptions and intermediate values allowed, an instruction to consider cultural and linguistic nuance, and a constraint to respond with a numerical value only. French prompts were direct translations of the English prompts. Table 3 shows the English-prompt/French-text version.[3]

Table 3
Wording of the English Prompt on French Text, by Component

Component	Wording
Task	Please analyze the sentiment of the following French text and provide a single numerical rating according to this scale:
Scale, -1.0	Strong negative sentiment - highly critical, hostile, or pessimistic content
Scale, -0.5	Moderate negative sentiment - somewhat negative, disapproving, or concerned content
Scale, 0.0	Neutral sentiment - factual, balanced, or neither positive nor negative content
Scale, 0.5	Moderate positive sentiment - somewhat positive, approving, or optimistic content
Scale, 1.0	Strong positive sentiment - highly supportive, enthusiastic, or optimistic content
Intermediate values	You may also use values between these points for intermediate sentiment levels (e.g., -0.7, 0.3).
Instruction 1	First, carefully read and understand the text, considering cultural and linguistic nuances in French.
Instruction 2	Analyze the emotional tone, word choice, and overall message.
Instruction 3	Respond ONLY with a single numerical value between -1.0 and 1.0 that best represents the sentiment.
Instruction 4	Do not include ANY explanations, analysis, or additional text in your response.
Input	Here is the text to analyze: [sentence]

The generative models were queried through the `ellmer` R package. Claude Haiku 4.5, Gemini 3.5 Flash, and GPT-5.6 Luna were accessed through their vendors' APIs. The ten

[3] The French-prompt/French-text version is identical except for language. See `src/93_prompts.R` in the project repository for the exact prompts, and `src/40_prompt.R` for the full evaluation pipeline, including model initialization, retry logic, rate limiting, and checkpointing.

open-weight models and Mistral Large 4 were accessed through OpenRouter. Responses were capped at 400 tokens. OpenRouter can serve the same model through several providers at different numerical precisions. Each model accessed through OpenRouter was therefore pinned to a single named provider endpoint, at the highest precision offered, with fallback to other providers disabled.[4] Gemini 3.5 Flash was queried with its default settings. Under these settings it generates reasoning tokens before answering: 354 output tokens per call on average, against 7 for Claude Haiku 4.5. Jev was queried through TypeSafe's API, with the model version pinned.[5] Each sentence was scored three times per model and condition (27,000 scores in total), and the mean of the three runs was taken as the final score. Despite explicit instructions to respond with a bare number, the generative models frequently returned formatted text, such as "Sentiment: 0.5". These responses were parsed with a pattern-matching function, and only values within the -1.0 to 1.0 range were retained. Jev's answers needed no parsing.

To check whether the sentence-level findings hold at the scale of a real corpus, the best-performing model on the benchmark, GPT-5.6 Luna, also scored all 2,683 full articles under the FR→FR and EN→FR conditions, with the same prompts and settings.[6] These scores are compared with the French Lexicoder dictionary in the Results (Figure 1).

Validation and Performance Evaluation

Performance was assessed with four metrics: Pearson correlation between each model's continuous predictions and the ground truth; mean absolute error (MAE); weighted F1 on a 7-category classification, from "very negative" to "very positive"; and weighted F1 on a 3-category classification (negative, neutral, positive) that groups the 7 categories into polarity bands. Weighted rather than macro F1 is used because the manually coded categories are

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[4] Endpoint pins (Mistral Large 4 is served only by Mistral itself), and the reasoning settings used to suppress chain-of-thought in models that reason by default, are listed in `src/94_models_map.R`. Mistral Large 4 reasons by default, exhausts the 400-token cap, and returns no answer. Its reasoning was disabled, as for Qwen3 32B and DeepSeek V3.2.

[5] See `src/42_jev_prompt.R` and `src/98_jev_helpers.R`.

[6] See `src/70_prompt_corpus.R`. The two passes required 5,366 API calls, 7.1 million input tokens and 187,000 output tokens, for USD 1.64 at list prices.

unevenly distributed. "Somewhat positive" alone accounts for 44% of annotated sentences, while "very negative" and "very positive" each account for 1% or less. Comparing the two classification schemes separates fine-grained discrimination (7 categories) from polarity classification (3 categories). All metrics were calculated on complete cases only.

Condition comparisons are paired by model and map onto the hypotheses. The prompt-language hypotheses compare FR→FR with EN→FR, which hold the French text constant. The translation hypotheses compare EN→FR with EN→EN, which hold the English prompt constant and isolate the effect of analyzing the translation. The comparison of FR→FR with EN→EN contrasts the two complete workflows a researcher might adopt. A non-significant t-test cannot by itself establish the absence of an effect. Alongside paired t-tests, two one-sided tests (TOST) therefore assess whether a difference is small enough to be treated as negligible, with an equivalence bound of ±0.05 on each metric's scale (Lakens, 2017). These comparisons and the averages reported in the Results cover the fourteen models that scored all 200 sentences (42 model-condition combinations). They exclude the dictionary baselines, which have no prompt, and Llama 3.2 1B, which scored a different partial subset of sentences in each condition.

Results

Prompt Language (H1a vs. H1b)

Holding the text constant, prompt language made no detectable difference (Figure 2 shows the three conditions for each model). French prompts on French text (FR→FR) scored slightly higher than English prompts on the same text (EN→FR) on every metric: mean correlation 0.708 vs. 0.687, MAE 0.208 vs. 0.216, 3-category F1 0.674 vs. 0.656, and 7-category F1 0.381 vs. 0.363. None of these paired differences was statistically significant (all $p > 0.10$). Equivalence tests go further. FR→FR and EN→FR were statistically equivalent within ±0.05 on MAE ($p < 0.001$), 7-category F1 ($p = 0.015$), and 3-category F1 ($p = 0.006$). Correlation was the one metric on which equivalence could not be established. The 90% confidence interval of the difference, [−0.024, 0.066], leaves room for a French-prompt advantage of up to 0.066, but not for an English-prompt advantage larger than 0.024. The results contradict H1a and support H1b on three

of four metrics.

The result holds at the scale of the corpus. Figure 1 plots, for each of the 2,671 full articles with all three scores, the score GPT-5.6 Luna gave under the English prompt against the score it gave under the French prompt (left), and the French Lexicoder score against the French-prompt score (right). The two prompt languages agree almost perfectly (r = 0.97; 94% of articles received the same sign). The dictionary and the model rank the articles very differently (r = 0.36 across articles, 0.42 across monthly means), and on 27% of articles they disagree even on the sign.[7] The comparison cannot say which of the two is right, since neither is the ground truth; the benchmark results below can.

Machine Translation (H2a vs. H2b)

For the LLMs, translation carried a cost. Holding the English prompt constant, analyzing the translation instead of the original (EN→EN vs. EN→FR) lowered 3-category F1 by 0.035 (0.621 vs. 0.656, $p = 0.004$). The accompanying drop in correlation (0.659 vs. 0.687) was not statistically significant ($p = 0.12$). Differences in MAE and 7-category F1 remained within the ±0.05 equivalence bound. Compared with the fully French workflow (FR→FR), the fully English workflow (EN→EN) lowered correlation by 0.048 ($p = 0.002$) and 3-category F1 by 0.053 ($p < 0.001$).

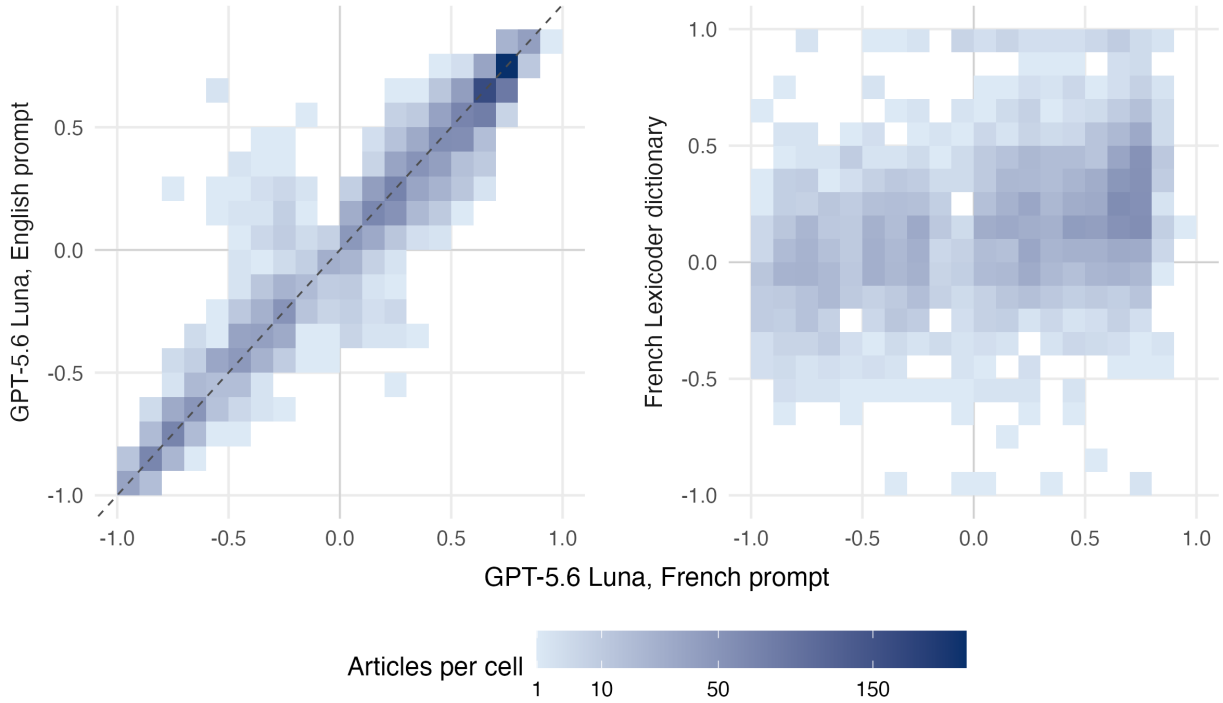
The dictionaries showed the opposite pattern. The English Lexicoder applied to the translation correlated with human judgments at r = 0.548, against 0.357 for the French Lexicoder applied to the original. The difference is 0.191 (95% bootstrap CI [0.059, 0.326]). The English dictionary also reached a higher 3-category F1 (0.588 vs. 0.557). The two dictionaries also differ in construction, so this contrast does not isolate language. Its direction nonetheless matches the bridge view. For the LLMs, the results support H2b: translation lowered polarity classification whichever prompt was held constant, and it lowered correlation when the two complete workflows are compared.

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[7] Correlation ignores differences of scale, so this is a disagreement about which articles are more positive than others, not about units.

Figure 1

Agreement Between Scoring Methods on the Full Corpus, by Article



Note. Each cell covers a 0.1 by 0.1 range of scores and is shaded by the number of articles it contains (n = 2,671). The horizontal axis is the GPT-5.6 Luna score under the French prompt. Left: the same model's score under the English prompt, with the identity line dashed (r = 0.97; the two prompts give the same sign on 94% of articles). Right: the French Lexicoder score on the original text (r = 0.36; the dictionary and the model give the same sign on 66% of articles and opposite signs on 28%, the remainder being articles that one of the two scores exactly zero). All scores run from −1 (very negative) to 1 (very positive).

Viability Against Dictionary Methods (RQ2)

General-purpose LLMs, used without fine-tuning, correlated with human judgments well above both dictionaries (Figure 2; exact values and confidence intervals in Table A3). The best configuration, GPT-5.6 Luna with French prompts, reached r = 0.81. Jev came close behind the leaders, with r = 0.77 in both conditions on French text and the second-highest 7-category F1 (0.528 in EN→FR). Twelve of the 42 model-condition combinations correlated with the ground truth more strongly than individual human coders agreed with the mean of the other two (mean

leave-one-out r = 0.748, range 0.724–0.779).[8]

Closed-weight models outperformed open-weight ones on average (mean r = 0.754 vs. 0.646; 3-category F1 0.685 vs. 0.632). The gap closed at the top of the open-weight range. Qwen3 235B-A22B reached r = 0.77 in FR→FR, above Gemini 3.5 Flash in the same condition. GPT-OSS 120B and GPT-OSS 20B averaged r = 0.73 and 0.72 across conditions. Mistral Small 3.2 reached the highest 3-category F1 of any open-weight model (0.721 in FR→FR), although its scores sat further from the human ratings (MAE 0.263 to 0.294). Mistral's newest flagship did not lead its own family: Mistral Large 4 scored below Mistral Small 3.2 on correlation (r = 0.69 to 0.71) and 3-category F1 (0.61 to 0.62), though with a lower MAE. Performance degraded at smaller sizes. Llama 3.1 8B remained usable (r = 0.58 to 0.61). Llama 3.2 3B was erratic (r = 0.48 in FR→FR but 0.13 in EN→FR). Llama 3.2 1B failed: on the 58 to 74 sentences it managed to score per condition, its correlation with the ground truth ranged from −0.20 to 0.10.

Cost was not a barrier. At list prices, scoring 1,000 sentences cost between USD 0.02 and 0.40 for every model but one (Table A1). The exception, Gemini 3.5 Flash (USD 3.77), reflects the reasoning tokens it generates by default rather than its per-token price.[9]

Against the LLM mean of r = 0.685, the dictionaries trailed on correlation (0.548 and 0.357) and on 3-category F1 (0.588 and 0.557 vs. 0.651). On correlation, the comparison is one-sided: every model from 8B parameters upward exceeded both dictionaries in every condition, and only Llama 3.2 3B and 1B fell below them. On 3-category F1, the translated English dictionary outperformed Qwen3 32B in every condition, GPT-OSS 120B in EN→EN, and the two smallest Llama models. On MAE, both dictionaries (0.203 and 0.210) did slightly better than the LLM average (0.211; Figure 2, right panel), because their scores cluster near zero on a sample dominated by mildly positive sentences.

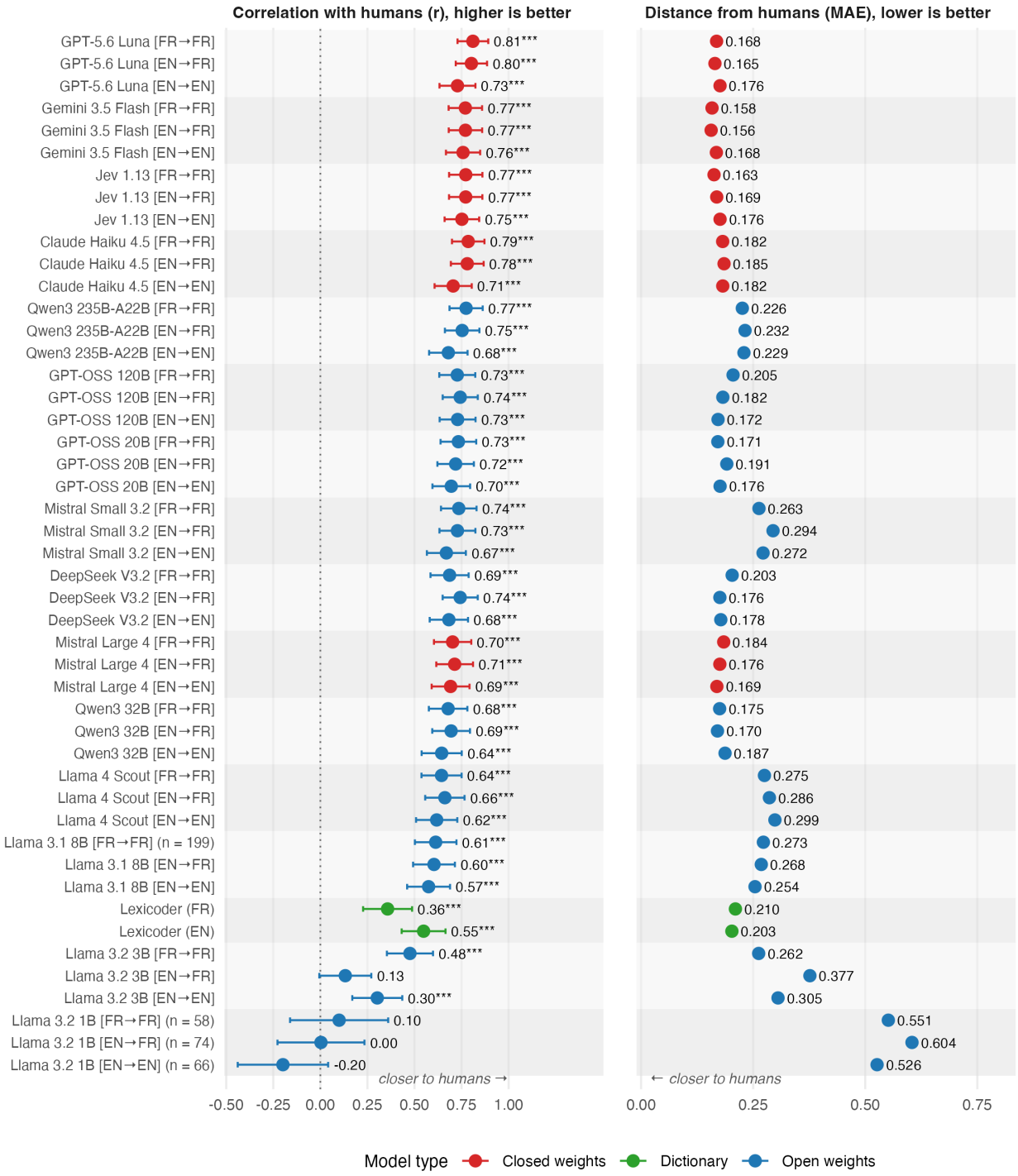
---

[8] This comparison slightly favors the models, because the ground truth they are compared against averages three coders rather than two. Agreement between annotators is not a strict ceiling on machine performance (Richie et al., 2022).

[9] No cost is reported for Llama 3.2 1B, whose endpoint returned the same input token count on every call; it also needed 22.7 API calls per usable score, against 1.0 to 1.03 for every other model.

Figure 2

Correlation and Mean Absolute Error Between Model Scores and Human Judgments, by Model and Condition



Note. One row per model and condition, sorted by each model's mean correlation and colored by model type. Left panel: Pearson correlation between the model's scores and the mean of three human coders, with 95% confidence intervals. Right panel: mean absolute error, the average distance between the model's score and the human score on the −1 to 1 scale, so that 0.17 means the model is typically 0.17 points away from the human rating. Within each model, the conditions are FR→FR (French prompt on French text), EN→FR (English prompt on French text), and EN→EN (English prompt on translated text). $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Classification Granularity

All models performed substantially better on 3-category than on 7-category classification (best F1 0.778 vs. 0.537; mean 0.651 vs. 0.369; Figure 3). Models identified positive and negative sentiment most reliably (7-category F1 = 0.480 and 0.454, rising to 0.706 and 0.626 in the 3-category scheme). They struggled most with the extreme categories (very negative F1 = 0.260, very positive F1 = 0.135), which are nearly absent from the sample (two sentences and one sentence). Neutral remained the hardest 3-category label (F1 = 0.272).

Discussion

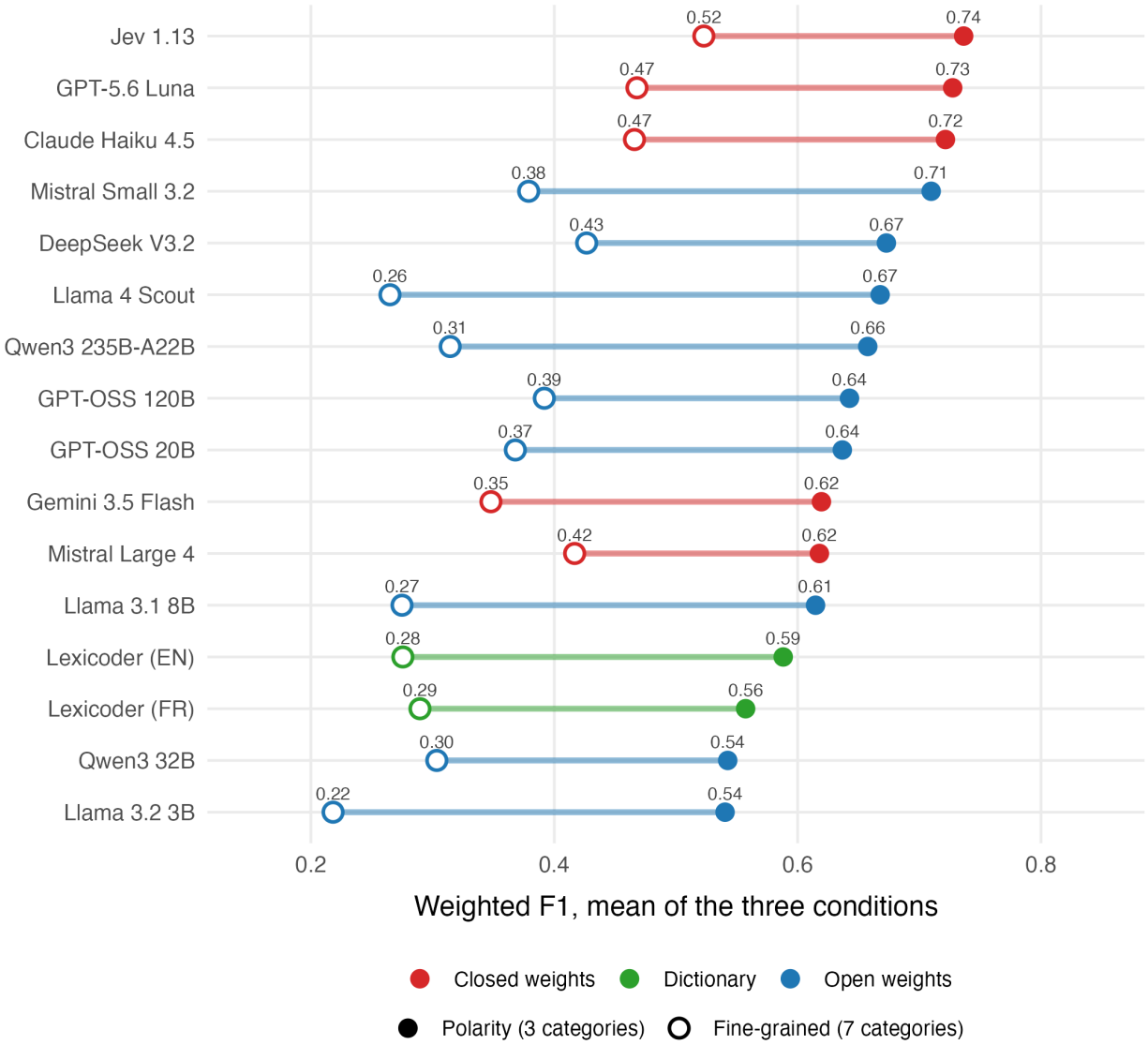
Prompt Language: Multilingual Competence, Within Scope Conditions

For French and current models, the results side with the multilingual-competence view. English prompts brought no advantage, and equivalence tests ruled out any English-prompt gain larger than a few hundredths, even on correlation, where strict equivalence could not be established. The equivalence held at the scale of the full corpus (Figure 1). Earlier social science evidence showed that labels agree across prompt languages (Rathje et al., 2024). This study extends that finding to the accuracy of the labels against human judgment. It is also consistent with the minimal differences observed for the strongest model in Arabic (Kmainasi et al., 2025). The results do not refute the pivot-language evidence so much as delimit it. That evidence comes largely from reasoning and inference benchmarks, older or smaller models, and lower-resourced languages (Lai et al., 2023; X. V. Lin et al., 2022; Muennighoff et al., 2023). The representational bias toward English that it points to (Wendler et al., 2024) need not produce worse instruction-following for a simple judgment in a language as well represented as French. The smaller open-weight models behaved differently. Llama 3.2 3B correlated with human judgments at r = 0.48 with French prompts but 0.13 with English ones. This suggests that sensitivity to prompt language may reappear below a certain model size, although a single model cannot establish this.

Prompt language does carry a cost that the accuracy results do not show. The French prompt is 6% longer than the English one in characters, yet it cost 10% to 34% more input tokens,

Figure 3

Weighted F1 on Polarity and on Fine-Grained Classification, by Model



Note. One row per model, from the highest polarity F1 to the lowest, colored by model type. Filled circles are the weighted F1 on three categories, hollow circles on seven, each averaged over the three conditions; the segment is the loss from grouping sentiment into seven categories instead of three. Llama 3.2 1B is omitted. Per-condition values are in the Appendix.

depending on the model (Table A2).[10] The difference is a property of each model's tokenizer, not of its size or license. Vocabularies that favor English encode English text with similar efficiency across models (1.17 to 1.25 tokens per word) but split French words into more pieces (1.38 to 1.87 tokens per word). Ahia et al. (2023) document the same disparity across many languages. Mistral's claim of a tokenizer that is cheaper on French (Mistral AI, 2024) held for one of its two models. Mistral Large 4 encoded French more compactly than any other model and had the smallest French-prompt overhead (10%). Mistral Small 3.2 did not stand out (1.45 tokens per French word, a 15% overhead). Output length did not differ by prompt language. Since accuracy was equivalent, the English prompt is the cheaper of two equally good options. The saving per 1,000 sentences ranges from under 6% for models whose output tokens weigh most in the bill to 10% to 31% for models billed mostly on input. Translating the text first would save a few percent more, at the accuracy cost reported above.

Translation: A Bridge for Dictionaries, a Loss for LLMs

The translation results are the article's main contribution to the debate. The same machine translation that improved the dictionary lowered the LLMs' performance. This divergence reconciles the bridge and loss views rather than choosing between them. Translation helps tools that cannot read the source language, and it costs information for tools that can. The dictionary result is consistent with a long line of evidence that translating into English lets researchers use better-validated English resources (De Vries et al., 2018; Proksch et al., 2019). LLMs already read French. For them, translation offers no such gain, and it exposes the analysis to the alterations of affective meaning that translation introduces (Mohammad et al., 2016). The same pattern appears, in passing, in the largest LLM study of party manifestos to date: inserting a machine-translation step before the analysis of 15 non-English manifestos lowered the correlation with expert surveys, and the authors dropped the step from their final design (Benoit et al., 2026). The present study makes that incidental observation the object of a formal test. The finding runs

[10] Jev's input is TypeSafe's structured question rather than the prompt shown above, so its token counts do not measure the same text. It is left out of the tokenizer comparisons in this paragraph.

against evidence that translating inputs into English improves multilingual models (Etxaniz et al., 2024). That evidence comes from benchmarks translated out of English, whereas the sentences analyzed here were written in French.

Viability: Optimism, With Validation

On RQ2, the results side with the optimistic literature (Gilardi et al., 2023; Rathje et al., 2024; T"ornberg, 2025) and extend it in two directions. First, the advantage of LLMs over dictionaries holds against a native-language dictionary as well as a translated one. Second, open-weight models were viable from about 8B parameters, and the largest approached the best closed-weight models. This is consistent with evidence from German (Klähn et al., 2025). Below that size, performance became erratic, and a 1B-parameter model could not follow the task. For researchers weighing the reproducibility concerns raised by proprietary models (Spirling, 2023), this makes a credible open-weight option available at low cost. Qwen3 235B-A22B and GPT-OSS 120B reached 90–95% of the best closed-weight correlation for USD 0.02–0.03 per thousand sentences. Models from a French developer showed no French advantage. Neither Mistral model did notably better with French prompts, and both trailed the leading closed-weight models. A different kind of model also proved viable. Jev, which returns a probability for each point of the scale rather than generated text, came within 0.04 of the best correlation for about USD 0.02 per thousand sentences. For French text, the answer to RQ2 is therefore yes. A general-purpose LLM measures sentiment better than the Lexicoder dictionary on every criterion that rewards discrimination, and the cost of running one is no longer a reason to prefer the dictionary. This is the condition under which comparative research in a non-English language becomes cheap: no translation step, no training data, no fine-tuning, a validated model, and a prompt in whichever language the researcher writes most easily.

The cautious literature is not dismissed. Performance varied widely across models, which reinforces the need to validate each model on each task (Pangakis et al., 2023). Two further results qualify the picture. First, the dictionaries' near-zero output gave them a slightly lower MAE than the LLMs. MAE alone can thus reward a method that barely discriminates. Second, all models

performed markedly worse on 7-category classification than on 3-category polarity. The two results fit together once one separates ranking sentences from placing them on the scale. LLMs order sentences and reproduce the spread of human ratings. Their correlation with the ground truth far exceeds that of the dictionaries. From 8B parameters upward, their scores spread at least as widely as the human ratings (SD 0.30 to 0.49 vs. 0.286), whereas the dictionaries' scores cluster near zero (SD = 0.065 for the English dictionary). What LLMs do poorly is place scores on the right side of fixed category thresholds. They overstate extremes: 10.6% of LLM scores exceed ±0.66, against 1.5% of ground-truth values. A sentence rated 0.5 by humans and 0.8 by a model is correctly ordered but misclassified. The ground truth contributes to the mismatch, since averaging three coders pulls it toward the center. Gradations in sentiment are one of the main arguments for dictionaries (Van Der Veen & Bleich, 2025). In this corpus, a single off-the-shelf dictionary captured them less well than the LLMs. The gap does not close with aggregation: on the full corpus, the dictionary agreed only weakly with the model, by article and by month (Figure 1). What the dictionary retains is transparency, since every score can be traced to the words that produced it, and stability over time, since a dictionary is not deprecated. These make it a useful baseline and robustness check alongside an LLM. They are not a reason to prefer it as the primary measure.

Limitations

Several limitations qualify these findings. The benchmark is modest in size and reliability. Its 200 sentences meet common size recommendations, but agreement between individual coders was moderate (Krippendorff's alpha = 0.597). Simulations show that such conditions raise the risk of misjudging automated methods (Song et al., 2020). Only two sentences fall in the very negative category and one in the very positive, so class-level F1 scores for these categories are unstable. The findings concern French only. The corpus is French-language news on a single domain, open-source software. Neither the absence of a prompt-language effect nor the cost of translation can be assumed to extend to other languages or genres.

Methodologically, the EN→EN condition relies on Google Translate, and translation

quality was deliberately not validated. The condition reproduces what a researcher who does not read the source language can do: translate automatically and analyze the result. Its lower performance should be read as the cost of that workflow rather than of translation as such. A single standardized prompt was used across the generative models for comparability. This may understate the performance achievable with model-specific prompting. Jev received the same scale in a different format, and its score is a probability-weighted average, which could in principle pull scores toward the middle of the scale and flatter its MAE. In practice its scores spread as widely as those of the best generative models (SD 0.36 against 0.37 for GPT-5.6 Luna; 7% of scores beyond ±0.66 against 6% to 11%), so its MAE is not an artefact of a compressed scale. Mistral Large 4 was evaluated as a preview release, which may change before its general availability. The models tested do not include fine-tuned or domain-specific classifiers, which can rival general-purpose LLMs (Mochtak et al., 2025). Costs are reported at list prices and count cached prompt tokens at the full input rate, so they are upper bounds. Finally, this evaluation is a snapshot. Several models from the earlier version of this study, including GPT-4o and Gemini 2.0 Flash, were deprecated by their providers within eighteen months, and the specific rankings should be expected to age quickly.

Conclusion

Researchers analyzing French text with LLMs can set aside both inherited reflexes. The language of the prompt is a matter of convenience. English instructions brought no advantage, and equivalence tests rule out any substantial one. The language of the text is not a matter of convenience. Translating into English improved the dictionary but lowered the LLMs' performance. The comparative strategy that served the dictionary era does not transfer to models that read the source language.

Two questions follow. The first is whether these results hold beyond French, particularly for languages less represented in training data, where the case for English prompts and translation is strongest. The design ports directly to such tests: the same three conditions, a manually annotated benchmark, and a different target language. The second is whether the size threshold

observed here, below which open-weight models become erratic and prompt-sensitive, is a stable feature of current models.

The practical answer is equally clear, and it matters most to the researchers the English-centered era left aside: those who work in another language, have no annotated training set, and cannot train or fine-tune a model. For them the choice is between an off-the-shelf dictionary and an off-the-shelf model, and the model wins. A general-purpose LLM, validated on a manually annotated sample, measures sentiment better than the Lexicoder dictionary on every criterion but transparency, and open-weight models from about 8B parameters do so for a few cents per thousand sentences. The dictionary remains a baseline and a robustness check. For researchers working with French text, the evidence settles three points: use a validated LLM, prompt in whichever language is convenient, and analyze the text in the language in which it was written.

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Author Contributions

Removed for anonymous review.

Statements and Declarations

Ethical Considerations

This study involved no human participants and collected no personal data. The 200 benchmark sentences were coded by members of the research team. Ethics approval was therefore not required.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Declaration of Conflicting Interests

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Data Availability

The code, the model scores, the manually annotated benchmark, and the prompts are available in a public repository (link withheld for anonymous review). The full texts of the news articles were obtained from the Eureka database under license and cannot be redistributed; the article identifiers needed to retrieve them are provided in the repository.

References

Ahia, O., Kumar, S., Gonen, H., Kasai, J., Mortensen, D., Smith, N., & Tsvetkov, Y. (2023). Do All Languages Cost the Same? Tokenization in the Era of Commercial Language Models. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 9904–9923. https://doi.org/10.18653/v1/2023.emnlp-main.614

Ahuja, K., Diddee, H., Hada, R., Ochieng, M., Ramesh, K., Jain, P., Nambi, A., Ganu, T., Segal, S., Ahmed, M., Bali, K., & Sitaram, S. (2023). MEGA: Multilingual Evaluation of Generative AI. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 4232–4267. https://doi.org/10.18653/v1/2023.emnlp-main.258

Baden, C., Pipal, C., Schoonvelde, M., & Van Der Velden, M. A. C. G. (2022). Three Gaps in Computational Text Analysis Methods for Social Sciences: A Research Agenda. *Communication Methods and Measures*, *16*(1), 1–18. https://doi.org/10.1080/19312458.2021.2015574

Benoit, K., De Marchi, S., Laver, C., Laver, M., & Ma, J. (2026). Using large language models to analyze political texts through natural language understanding. *American Journal of*

Political Science. https://doi.org/10.1111/ajps.70050

De Vries, E., Schoonvelde, M., & Schumacher, G. (2018). No Longer Lost in Translation: Evidence that Google Translate Works for Comparative Bag-of-Words Text Applications. *Political Analysis*, *26*(4), 417–430. https://doi.org/10.1017/pan.2018.26

Duval, D., & Pétry, F. (2016). L'analyse automatisée du ton médiatique : Construction et utilisation de la version française du Lexicoder Sentiment Dictionary. *Canadian Journal of Political Science/Revue Canadienne de Science Politique*, *49*(2), 197–220. https://doi.org/10.1017/S000842391600055X

Etxaniz, J., Azkune, G., Soroa, A., Lopez de Lacalle, O., & Artetxe, M. (2024). Do multilingual language models think better in English? *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, 550–564. https://doi.org/10.18653/v1/2024.naacl-short.46

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, *120*(30), e2305016120. https://doi.org/10.1073/pnas.2305016120

Grimmer, J., & Stewart, B. M. (2013). Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts. *Political Analysis*, *21*(3), 267–297. https://doi.org/10.1093/pan/mps028

Heseltine, M., & Clemm Von Hohenberg, B. (2024). Large language models as a substitute for human experts in annotating political text. *Research & Politics*, *11*(1), 20531680241236239. https://doi.org/10.1177/20531680241236239

Klähn, J., Borst-Graetz, J., & Burghardt, M. (2025). From dictionaries to LLMs – an evaluation of sentiment analysis techniques for German language data. *Computational Humanities Research*, *1*, e4. https://doi.org/10.1017/chr.2025.10005

Kmainasi, M. B., Khan, R., Shahroor, A. E., Bendou, B., Hasanain, M., & Alam, F. (2025). Native vs non-native language prompting: A comparative analysis. *Web Information*

Systems Engineering – WISE 2024. https://doi.org/10.1007/978-981-96-0576-7_30

Lai, V., Ngo, N., Pouran Ben Veyseh, A., Man, H., Dernoncourt, F., Bui, T., & Nguyen, T. (2023). ChatGPT Beyond English: Towards a Comprehensive Evaluation of Large Language Models in Multilingual Learning. *Findings of the Association for Computational Linguistics: EMNLP 2023*, 13171–13189. https://doi.org/10.18653/v1/2023.findings-emnlp.878

Lakens, D. (2017). Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science*, *8*(4), 355–362. https://doi.org/10.1177/1948550617697177

Laurer, M., Van Atteveldt, W., Casas, A., & Welbers, K. (2024). Less Annotating, More Classifying: Addressing the Data Scarcity Issue of Supervised Machine Learning with Deep Transfer Learning and BERT-NLI. *Political Analysis*, *32*(1), 84–100. https://doi.org/10.1017/pan.2023.20

Licht, H. (2023). Cross-Lingual Classification of Political Texts Using Multilingual Sentence Embeddings. *Political Analysis*, *31*(3), 366–379. https://doi.org/10.1017/pan.2022.29

Lin, H., & Zhang, Y. (2026). Navigating the Risks of Using Large Language Models for Text Annotation in Social Science Research. *Social Science Computer Review*, *44*(3), 403–427. https://doi.org/10.1177/08944393251366243

Lin, X. V., Mihaylov, T., Artetxe, M., Wang, T., Chen, S., Simig, D., Ott, M., Goyal, N., Bhosale, S., Du, J., Pasunuru, R., Shleifer, S., Koura, P. S., Chaudhary, V., O'Horo, B., Wang, J., Zettlemoyer, L., Kozareva, Z., Diab, M., … Li, X. (2022). Few-shot Learning with Multilingual Generative Language Models. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. https://doi.org/10.18653/v1/2022.emnlp-main.616

Lucas, C., Nielsen, R. A., Roberts, M. E., Stewart, B. M., Storer, A., & Tingley, D. (2015). Computer-Assisted Text Analysis for Comparative Politics. *Political Analysis*, *23*(2), 254–277. https://doi.org/10.1093/pan/mpu019

Mistral AI. (2024). *Mistral NeMo*. https://mistral.ai/news/mistral-nemo

Mochtak, M., Rupnik, P., Kuzman, T., & Ljubešić, N. (2025). Parlasent: Mapping sentiment in political discourse with large language models. *Political Research Exchange*, *7*(1), 2508377. https://doi.org/10.1080/2474736X.2025.2508377

Mohammad, S. M., Salameh, M., & Kiritchenko, S. (2016). How translation alters sentiment. *Journal of Artificial Intelligence Research*, *55*, 95–130. https://doi.org/10.1613/jair.4787

Muennighoff, N., Wang, T., Sutawika, L., Roberts, A., Biderman, S., Le Scao, T., Bari, M. S., Shen, S., Yong, Z. X., Schoelkopf, H., Tang, X., Radev, D., Aji, A. F., Almubarak, K., Albanie, S., Alyafeai, Z., Webson, A., Raff, E., & Raffel, C. (2023). Crosslingual Generalization through Multitask Finetuning. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. https://doi.org/10.18653/v1/2023.acl-long.891

Palmer, A., Smith, N. A., & Spirling, A. (2023). Using proprietary language models in academic research requires explicit justification. *Nature Computational Science*, *4*(1), 2–3. https://doi.org/10.1038/s43588-023-00585-1

Pangakis, N., Wolken, S., & Fasching, N. (2023). *Automated Annotation with Generative AI Requires Validation* (arXiv:2306.00176). arXiv. https://doi.org/10.48550/arXiv.2306.00176

Proksch, S.-O., Lowe, W., Wäckerle, J., & Soroka, S. (2019). Multilingual Sentiment Analysis: A New Approach to Measuring Conflict in Legislative Speeches. *Legislative Studies Quarterly*, *44*(1), 97–131. https://doi.org/10.1111/lsq.12218

Rathje, S., Mirea, D.-M., Sucholutsky, I., Marjieh, R., Robertson, C. E., & Van Bavel, J. J. (2024). GPT is an effective tool for multilingual psychological text analysis. *Proceedings of the National Academy of Sciences*, *121*(34), e2308950121. https://doi.org/10.1073/pnas.2308950121

Richie, R., Grover, S., & Tsui, F. (2022). Inter-annotator agreement is not the ceiling of machine learning performance: Evidence from a comprehensive set of simulations. *Proceedings of*

the 21st Workshop on Biomedical Language Processing, 275–284. https://doi.org/10.18653/v1/2022.bionlp-1.26

Shi, F., Suzgun, M., Freitag, M., Wang, X., Srivats, S., Vosoughi, S., Chung, H. W., Tay, Y., Ruder, S., Zhou, D., Das, D., & Wei, J. (2022). *Language Models are Multilingual Chain-of-Thought Reasoners* (arXiv:2210.03057). arXiv. https://doi.org/10.48550/arXiv.2210.03057

Song, H., Tolochko, P., Eberl, J.-M., Eisele, O., Greussing, E., Heidenreich, T., Lind, F., Galyga, S., & Boomgaarden, H. G. (2020). In Validations We Trust? The Impact of Imperfect Human Annotations as a Gold Standard on the Quality of Validation of Automated Content Analysis. *Political Communication*, *37*(4), 550–572. https://doi.org/10.1080/10584609.2020.1723752

Spirling, A. (2023). Why open-source generative AI models are an ethical way forward for science. *Nature*, *616*, 413. https://doi.org/10.1038/d41586-023-01295-4

T"ornberg, P. (2025). Large language models outperform expert coders and supervised classifiers at annotating political social media messages. *Social Science Computer Review*, *43*(6), 1181–1195. https://doi.org/10.1177/08944393241286471

Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A., Grave, E., & Lample, G. (2023). *LLaMA: Open and Efficient Foundation Language Models* (arXiv:2302.13971). arXiv. https://arxiv.org/abs/2302.13971

van Atteveldt, W., van der Velden, M. A. C. G., & Boukes, M. (2021). The validity of sentiment analysis: Comparing manual annotation, crowd-coding, dictionary approaches, and machine learning algorithms. *Communication Methods and Measures*, *15*(2), 121–140. https://doi.org/10.1080/19312458.2020.1869198

Van Der Veen, A. M., & Bleich, E. (2025). The advantages of lexicon-based sentiment analysis in an age of machine learning. *PLOS ONE*, *20*(1), e0313092. https://doi.org/10.1371/journal.pone.0313092

Wendler, C., Veselovsky, V., Monea, G., & West, R. (2024). Do Llamas Work in English? On the Latent Language of Multilingual Transformers. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 15366–15394. https://doi.org/10.18653/v1/2024.acl-long.820

Widmann, T., & Wich, M. (2023). Creating and Comparing Dictionary, Word Embedding, and Transformer-Based Models to Measure Discrete Emotions in German Political Text. *Political Analysis*, *31*(4), 626–641. https://doi.org/10.1017/pan.2022.15

Young, L., & Soroka, S. (2012). Affective News: The Automated Coding of Sentiment in Political Texts. *Political Communication*, *29*(2), 205–231. https://doi.org/10.1080/10584609.2012.671234

Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., & Yang, D. (2024). Can Large Language Models Transform Computational Social Science? *Computational Linguistics*, *50*(1), 237–291. https://doi.org/10.1162/coli_a_00502

Appendix

Additional Results

Table A1

Cost of Scoring 1,000 Sentences at List Prices (Upper Bound; Cached Prompt Tokens Counted at the Full Input Rate; Input Tokens and Cost Not Reported for Llama 3.2 1B, Whose Endpoint Returned a Constant Token Count)

Model	Provider	Weights	API calls per score	Input tokens	Output tokens	USD per 1,000 sentences
GPT-OSS 20B	OpenRouter	Open	1.02	657,496	87,892	0.018
Llama 3.2 3B	OpenRouter	Open	1.00	595,142	7,320	0.018
Jev 1.13	TypeSafe	Closed	1.00	815,199	32,400	0.019
GPT-OSS 120B	OpenRouter	Open	1.00	667,425	95,660	0.020
Qwen3 32B	OpenRouter	Open	1.00	592,086	5,696	0.027
Mistral Small 3.2	OpenRouter	Open	1.00	553,761	7,405	0.029
Qwen3 235B-A22B	OpenRouter	Open	1.00	584,886	7,752	0.030
Llama 4 Scout	OpenRouter	Open	1.00	564,947	8,467	0.059
DeepSeek V3.2	OpenRouter	Open	1.00	559,594	9,340	0.067
Llama 3.1 8B	OpenRouter	Open	1.01	593,398	8,062	0.074
GPT-5.6 Luna	OpenAI	Closed	1.00	552,225	51,819	0.096
Mistral Large 4	OpenRouter	Closed	1.00	534,990	7,603	0.211
Claude Haiku 4.5	Anthropic	Closed	1.00	648,705	12,600	0.395
Gemini 3.5 Flash	Google	Closed	1.03	568,160	658,571	3.766
Llama 3.2 1B	OpenRouter	Open	22.68	–	168,789	–

Table A2

Mean Input Tokens per API Call by Language Condition, and Tokens per Word by Text Language (Valid Calls Only; Llama 3.2 1B Excluded Because Its Endpoint Reported a Constant Token Count on Every Call; Jev's Input Is a Structured Question Rather Than the Text Prompt, So Its Counts Are Not Comparable)

Model	EN→EN	EN→FR	FR→FR	French-prompt overhead	Tokens per French word	Tokens per English word
Mistral Large 4	284.5	289.1	318.1	+29 (+10.0%)	1.38	1.22
Jev 1.13	431.0	440.3	487.3	+47 (+10.7%)	1.62	1.22
GPT-OSS 120B	352.0	357.7	402.7	+45 (+12.6%)	1.42	1.19
GPT-OSS 20B	339.0	344.7	389.7	+45 (+13.1%)	1.42	1.19
Llama 4 Scout	294.1	299.9	343.9	+44 (+14.7%)	1.41	1.19
Mistral Small 3.2	288.6	294.6	339.6	+45 (+15.3%)	1.45	1.22
GPT-5.6 Luna	288.0	293.7	338.7	+45 (+15.3%)	1.42	1.19
Gemini 3.5 Flash	285.0	292.1	337.1	+45 (+15.4%)	1.46	1.17
Qwen3 32B	296.5	307.7	382.7	+75 (+24.4%)	1.69	1.21
DeepSeek V3.2	280.1	290.3	361.3	+71 (+24.5%)	1.61	1.19
Qwen3 235B-A22B	292.5	303.7	378.7	+75 (+24.7%)	1.69	1.21
Llama 3.2 3B	295.2	306.8	388.8	+82 (+26.7%)	1.69	1.20
Llama 3.1 8B	295.2	306.8	389.0	+82 (+26.8%)	1.69	1.20
Claude Haiku 4.5	313.8	328.2	439.2	+111 (+33.8%)	1.87	1.25

Table A3

Correlation With Human Judgments (With 95% Confidence Interval), Mean Absolute Error, and Number of Sentences Scored, by Model and Condition

Model	Weights	Condition	r	95% CI	MAE	n
GPT-5.6 Luna	Closed	FR→FR	0.810	[0.73, 0.89]	0.168	200
GPT-5.6 Luna	Closed	EN→FR	0.802	[0.72, 0.89]	0.165	200
GPT-5.6 Luna	Closed	EN→EN	0.728	[0.63, 0.82]	0.176	200
Gemini 3.5 Flash	Closed	FR→FR	0.770	[0.68, 0.86]	0.158	200
Gemini 3.5 Flash	Closed	EN→FR	0.771	[0.68, 0.86]	0.156	200
Gemini 3.5 Flash	Closed	EN→EN	0.758	[0.67, 0.85]	0.168	200
Jev 1.13	Closed	FR→FR	0.772	[0.68, 0.86]	0.163	200
Jev 1.13	Closed	EN→FR	0.772	[0.68, 0.86]	0.169	200
Jev 1.13	Closed	EN→EN	0.752	[0.66, 0.84]	0.176	200
Claude Haiku 4.5	Closed	FR→FR	0.785	[0.70, 0.87]	0.182	200
Claude Haiku 4.5	Closed	EN→FR	0.780	[0.69, 0.87]	0.185	200
Claude Haiku 4.5	Closed	EN→EN	0.705	[0.61, 0.80]	0.182	200
Qwen3 235B-A22B	Open	FR→FR	0.774	[0.69, 0.86]	0.226	200
Qwen3 235B-A22B	Open	EN→FR	0.753	[0.66, 0.84]	0.232	200
Qwen3 235B-A22B	Open	EN→EN	0.680	[0.58, 0.78]	0.229	200
GPT-OSS 120B	Open	FR→FR	0.727	[0.63, 0.82]	0.205	200
GPT-OSS 120B	Open	EN→FR	0.743	[0.65, 0.84]	0.182	200
GPT-OSS 120B	Open	EN→EN	0.729	[0.63, 0.82]	0.172	200
GPT-OSS 20B	Open	FR→FR	0.733	[0.64, 0.83]	0.171	200
GPT-OSS 20B	Open	EN→FR	0.718	[0.62, 0.82]	0.191	200
GPT-OSS 20B	Open	EN→EN	0.695	[0.60, 0.80]	0.176	200
Mistral Small 3.2	Open	FR→FR	0.735	[0.64, 0.83]	0.263	200
Mistral Small 3.2	Open	EN→FR	0.728	[0.63, 0.82]	0.294	200
Mistral Small 3.2	Open	EN→EN	0.669	[0.57, 0.77]	0.272	200
DeepSeek V3.2	Open	FR→FR	0.686	[0.58, 0.79]	0.203	200
DeepSeek V3.2	Open	EN→FR	0.743	[0.65, 0.84]	0.176	200
DeepSeek V3.2	Open	EN→EN	0.683	[0.58, 0.78]	0.178	200
Mistral Large 4	Closed	FR→FR	0.702	[0.60, 0.80]	0.184	200
Mistral Large 4	Closed	EN→FR	0.713	[0.62, 0.81]	0.176	200
Mistral Large 4	Closed	EN→EN	0.692	[0.59, 0.79]	0.169	200
Qwen3 32B	Open	FR→FR	0.679	[0.58, 0.78]	0.175	200
Qwen3 32B	Open	EN→FR	0.695	[0.59, 0.79]	0.170	200
Qwen3 32B	Open	EN→EN	0.644	[0.54, 0.75]	0.187	200
Llama 4 Scout	Open	FR→FR	0.644	[0.54, 0.75]	0.275	200
Llama 4 Scout	Open	EN→FR	0.661	[0.56, 0.77]	0.286	200
Llama 4 Scout	Open	EN→EN	0.618	[0.51, 0.73]	0.299	200
Llama 3.1 8B	Open	FR→FR	0.612	[0.50, 0.72]	0.273	199
Llama 3.1 8B	Open	EN→FR	0.604	[0.49, 0.71]	0.268	200
Llama 3.1 8B	Open	EN→EN	0.575	[0.46, 0.69]	0.254	200
Lexicoder (EN)	Dictionary	EN	0.548	[0.43, 0.66]	0.203	200
Lexicoder (FR)	Dictionary	FR	0.357	[0.23, 0.49]	0.210	200
Llama 3.2 3B	Open	FR→FR	0.476	[0.35, 0.60]	0.262	200
Llama 3.2 3B	Open	EN→FR	0.132	[-0.01, 0.27]	0.377	200
Llama 3.2 3B	Open	EN→EN	0.302	[0.17, 0.44]	0.305	200
Llama 3.2 1B	Open	FR→FR	0.100	[-0.16, 0.36]	0.551	58
Llama 3.2 1B	Open	EN→FR	0.003	[-0.23, 0.23]	0.604	74
Llama 3.2 1B	Open	EN→EN	-0.199	[-0.44, 0.04]	0.526	66

Table A4
F1 Scores for 3-Category Classification

Model	Negative	Neutral	Positive	Weighted F1
Claude Haiku 4.5 (FR→FR)	0.762	0.375	0.832	0.778
GPT-5.6 Luna (FR→FR)	0.748	0.385	0.800	0.755
Jev 1.13 (EN→FR)	0.691	0.500	0.810	0.751
Jev 1.13 (FR→FR)	0.695	0.400	0.803	0.741
GPT-5.6 Luna (EN→FR)	0.730	0.364	0.776	0.733
Mistral Small 3.2 (FR→FR)	0.578	0.286	0.844	0.721
Jev 1.13 (EN→EN)	0.711	0.222	0.777	0.717
Claude Haiku 4.5 (EN→FR)	0.690	0.273	0.778	0.715
Mistral Small 3.2 (EN→FR)	0.545	0.364	0.841	0.715
DeepSeek V3.2 (FR→FR)	0.672	0.216	0.765	0.697
GPT-5.6 Luna (EN→EN)	0.688	0.367	0.736	0.695
Qwen3 235B-A22B (FR→FR)	0.752	0.243	0.716	0.695
Mistral Small 3.2 (EN→EN)	0.488	0.364	0.836	0.693
DeepSeek V3.2 (EN→FR)	0.696	0.226	0.737	0.688
GPT-OSS 120B (FR→FR)	0.721	0.314	0.712	0.687
Llama 4 Scout (FR→FR)	0.643	0.291	0.747	0.682
Qwen3 235B-A22B (EN→FR)	0.719	0.254	0.706	0.679
Llama 4 Scout (EN→FR)	0.667	0.258	0.732	0.678
GPT-OSS 120B (EN→FR)	0.717	0.324	0.689	0.672
Claude Haiku 4.5 (EN→EN)	0.678	0.256	0.715	0.671
GPT-OSS 20B (EN→FR)	0.700	0.268	0.677	0.655
GPT-OSS 20B (FR→FR)	0.691	0.258	0.680	0.654
Llama 4 Scout (EN→EN)	0.647	0.246	0.686	0.643
Gemini 3.5 Flash (FR→FR)	0.655	0.264	0.673	0.639
DeepSeek V3.2 (EN→EN)	0.672	0.258	0.657	0.633
Mistral Large 4 (EN→EN)	0.604	0.250	0.677	0.624
Llama 3.1 8B (EN→FR)	0.652	0.281	0.646	0.623
Mistral Large 4 (FR→FR)	0.644	0.259	0.650	0.621
Llama 3.2 3B (FR→FR)	0.465	0.195	0.747	0.620
Llama 3.1 8B (EN→EN)	0.627	0.265	0.653	0.618
Gemini 3.5 Flash (EN→FR)	0.642	0.242	0.646	0.616
Mistral Large 4 (EN→FR)	0.648	0.235	0.632	0.609
Llama 3.1 8B (FR→FR)	0.615	0.174	0.643	0.604
Gemini 3.5 Flash (EN→EN)	0.667	0.233	0.613	0.603
GPT-OSS 20B (EN→EN)	0.614	0.234	0.635	0.601
Qwen3 235B-A22B (EN→EN)	0.698	0.227	0.591	0.600
Lexicoder (EN)	0.574	0.231	0.636	0.588
GPT-OSS 120B (EN→EN)	0.678	0.218	0.552	0.569
Qwen3 32B (EN→FR)	0.554	0.165	0.621	0.568
Lexicoder (FR)	0.474	0.289	0.631	0.557
Qwen3 32B (FR→FR)	0.515	0.177	0.611	0.550
Llama 3.2 3B (EN→EN)	0.167	0.261	0.730	0.520
Qwen3 32B (EN→EN)	0.559	0.190	0.520	0.509
Llama 3.2 3B (EN→FR)	0.000	0.222	0.757	0.481
Llama 3.2 1B (FR→FR)	0.000	0.138	0.648	0.443
Llama 3.2 1B (EN→FR)	0.000	0.000	0.568	0.338

Table A5

F1 Scores for 7-Category Classification

Model	Very neg.	Negative	Somewhat neg.	Neutral	Somewhat pos.	Positive	Very pos.	Weighted F1
Claude Haiku 4.5 (FR→FR)	0.400	0.680	0.455	0.375	0.540	0.612	0.000	0.537
Jev 1.13 (EN→FR)	0.200	0.586	0.282	0.500	0.623	0.548	0.250	0.528
Jev 1.13 (FR→FR)	0.222	0.621	0.324	0.400	0.601	0.540	0.250	0.523
Jev 1.13 (EN→EN)	0.200	0.525	0.385	0.222	0.610	0.566	0.286	0.518
GPT-5.6 Luna (FR→FR)	0.286	0.500	0.393	0.385	0.538	0.462	0.133	0.479
GPT-5.6 Luna (EN→EN)	0.364	0.500	0.338	0.367	0.510	0.542	0.250	0.470
GPT-5.6 Luna (EN→FR)	0.333	0.500	0.258	0.364	0.564	0.407	0.125	0.455
DeepSeek V3.2 (EN→EN)	0.333	0.533	0.419	0.258	0.415	0.588	0.667	0.448
Claude Haiku 4.5 (EN→EN)	0.286	0.642	0.400	0.256	0.409	0.568	0.000	0.448
GPT-OSS 120B (EN→FR)	0.333	0.566	0.400	0.324	0.416	0.485	0.133	0.434
Mistral Large 4 (EN→EN)	0.333	0.651	0.316	0.250	0.444	0.408	0.000	0.423
DeepSeek V3.2 (EN→FR)	0.444	0.667	0.429	0.226	0.349	0.493	0.000	0.416
Mistral Small 3.2 (EN→EN)	0.333	0.345	0.222	0.364	0.549	0.364	0.091	0.416
Mistral Large 4 (EN→FR)	0.167	0.600	0.357	0.235	0.361	0.590	0.286	0.416
DeepSeek V3.2 (FR→FR)	0.333	0.368	0.469	0.216	0.400	0.527	0.167	0.415
Claude Haiku 4.5 (EN→FR)	0.500	0.680	0.364	0.273	0.347	0.511	0.200	0.413
Mistral Large 4 (FR→FR)	0.154	0.488	0.344	0.259	0.397	0.561	0.222	0.412
Mistral Small 3.2 (FR→FR)	0.500	0.400	0.170	0.286	0.509	0.380	0.125	0.394
GPT-OSS 20B (EN→FR)	0.235	0.431	0.269	0.268	0.370	0.535	0.000	0.376
GPT-OSS 120B (EN→EN)	0.444	0.692	0.281	0.218	0.288	0.536	0.286	0.373
Gemini 3.5 Flash (FR→FR)	0.222	0.583	0.250	0.264	0.310	0.571	0.200	0.370
GPT-OSS 120B (FR→FR)	0.308	0.545	0.333	0.314	0.319	0.447	0.154	0.369
GPT-OSS 20B (FR→FR)	0.364	0.600	0.286	0.258	0.325	0.448	0.286	0.367
GPT-OSS 20B (EN→EN)	0.400	0.528	0.235	0.234	0.338	0.508	0.000	0.362
Gemini 3.5 Flash (EN→FR)	0.400	0.694	0.255	0.242	0.263	0.485	0.167	0.349
Qwen3 32B (EN→FR)	0.222	0.512	0.163	0.165	0.286	0.615	0.000	0.334
Qwen3 235B-A22B (FR→FR)	0.190	0.407	0.320	0.243	0.319	0.369	0.087	0.330
Mistral Small 3.2 (EN→FR)	0.364	0.312	0.133	0.364	0.420	0.292	0.062	0.326
Gemini 3.5 Flash (EN→EN)	0.364	0.652	0.222	0.233	0.230	0.500	0.000	0.324
Qwen3 32B (FR→FR)	0.333	0.558	0.125	0.177	0.236	0.632	0.000	0.315
Qwen3 235B-A22B (EN→FR)	0.182	0.321	0.240	0.254	0.350	0.339	0.091	0.315
Qwen3 235B-A22B (EN→EN)	0.182	0.275	0.302	0.227	0.279	0.408	0.133	0.299
Llama 3.1 8B (EN→FR)	0.111	0.120	0.192	0.281	0.319	0.507	0.200	0.297
Lexicoder (FR)	0.000	0.000	0.253	0.289	0.479	0.057	0.000	0.290
Llama 4 Scout (FR→FR)	0.125	0.167	0.182	0.291	0.341	0.356	0.100	0.286
Lexicoder (EN)	0.000	0.000	0.267	0.231	0.472	0.000	0.000	0.276
Llama 3.1 8B (FR→FR)	0.143	0.197	0.179	0.174	0.239	0.543	0.000	0.267
Llama 3.2 3B (FR→FR)	0.000	0.364	0.200	0.195	0.240	0.395	0.000	0.266
Qwen3 32B (EN→EN)	0.222	0.364	0.172	0.190	0.168	0.567	0.333	0.261
Llama 3.1 8B (EN→EN)	0.138	0.222	0.136	0.265	0.246	0.479	0.143	0.261
Llama 4 Scout (EN→FR)	0.154	0.261	0.143	0.258	0.281	0.329	0.080	0.257
Llama 4 Scout (EN→EN)	0.105	0.280	0.125	0.246	0.234	0.434	0.154	0.252
Llama 3.2 3B (EN→EN)	0.000	0.138	0.050	0.261	0.250	0.375	0.000	0.217
Llama 3.2 3B (EN→FR)	0.000	0.000	0.000	0.222	0.232	0.317	0.000	0.171
Llama 3.2 1B (EN→EN)	0.000	0.000	0.000	0.150	0.000	0.211	0.000	0.051
Llama 3.2 1B (FR→FR)	0.000	0.000	0.000	0.138	0.000	0.000	0.000	0.007
Llama 3.2 1B (EN→FR)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000